MIL-D-13796A(Ord)
14 May 1962
SUPERSEDING
MIL-D-13796(Ord)
18 November 1954

MILITARY SPECIFICATION

DI-2 ETHYLHEXYLPHTHALATE

1. SCOPE

1.1 This specification covers Di-2-ethylhexylphthalate for use in explosives and propellant.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-729 - Drums; Metal, 55 gallon (for Shipment of Noncorrosive Material).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 - Inspection Terms and Definitions.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-286 - Propellants, Sampling, Inspection and Testing.

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION PUBLICATION

National Motor Freight Classification Rules and Container Specifications

(Application for copies should be addressed to the American Trucking Association 1424 16th Street N. W. Washington, D. C.)

3. REQUIREMENTS

3.1 Properties. - The properties of Di-2-ethylhexylphthalate shall be in accordance with table I when determined as specified in the applicable paragraph.

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Table I

Property		Applicable paragraphs
1. Percent di-2-ethylhexylphthalate by weight minimum (min.)	99.0	4.3.1
2. Specific gravity at 20/20 degrees centigrade (°C.)	0.9840-0.9885	4.3.2
3. Acidity (as phthalic acid), percent maximum (max.)	0.01	4.3.3
4. Ash, percent max.	0.01	4.3.5

3.2 Color. -

3.2.1 Color as received. - The color shall not be darker than that of the 50-parts per million (p. p. m.) platinum cobalt standard, when determined as specified in 4.3.4.2.

3.2.2 Heat test color. - The heat test color shall not be darker than that of the 75 p. p. m. platinum cobalt standard, when determined as specified in 4.3.4.3.

3.3 Workmanship. - The standard of workmanship shall be such as to insure the production of material meeting the requirements of this specification. The Di-2-ethylhexylphthalate shall be free from suspended matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government

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as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system. -If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall

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notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product. -At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Quantity of product inspected.

c. Results obtained for all inspection performed.

d. Specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is covered by referenced Government specifications.

f. Quantity of product in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification. -Using the contractor's written quality assurance procedure (see 4.1.1), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 Surveillance. -Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

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c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection. -Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions. -

4.2.1 Lot formation. -A lot shall consist of one or more batches of Di-2-ethylhexylphthalate, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of Di-2-ethylhexylphthalate that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination. -Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans, in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity.

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4.2.2.1 Sealed container. -

Categories	Defects	Method of inspection
Critical:		
	1. Leak in containers	Visual
Major:	AQL 1.00 percent	
	101. Marking misleading or unidentifiable	Visual
	102. Lead seal missing or unsealed	Visual
Minor:	AQL 2.50 percent	
	201. Container not filled to correct volume . . .	Scale
	202. Evidence of poor workmanship.	Visual

4.2.3 Testing. -

4.2.3.1 Sampling. -

4.2.3.1.1 Containers. -A 32 ounce sample shall be selected at random from the lot by the inspector. The sample shall be thoroughly mixed, and placed in an airtight glass bottle, having a glass stopper or a cap containing a cork or cardboard liner covered with metal foil, and labeled to show the name of the material, manufacturer, plant, contract or purchase order number, and lot number. All acceptance tests shall be made on this sample. However, if by visual observation it becomes apparent during sampling that the batch is not uniform, the inspector may require that additional samples be taken and tested for compliance with the requirements of the specification.

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4.2.3.1.2 Tank cars. -By means of a clean, weighted, small-neck, glass bottle, approximately one quart of Di-2-ethylhexylphthalate shall be removed. The unstoppered bottle shall be rinsed with the material being sampled, and lowered to the bottom of the tank by means of a cord, chain, or rod. The bottle shall be immediately withdrawn to the surface. In order to obtain a representative cross-section of the material, the speed of lowering and raising the bottle shall be uniform, and so regulated that the bottle is just filled as it reaches the surface of the liquid. This sample shall be placed in a glass bottle, having a glass stopper or cap containing a cork or cardboard liner covered with metal foil, and labeled to show the name of the material, manufacturer, plant, contract or order number, and number of gallons in the lot. All acceptance tests shall be made on this sample. If any test of the sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures. -

4.3.1 Determination of Di-2-ethylhexylphthalate. -A 1.0 gram (gm.) portion of the sample shall be accurately weighed, and transferred to a 250 milliliter (ml.) Erlenmeyer flask having a standard taper ground glass neck. Approximately 25 ml. of 0.5 normal (N) alkali, accurately measured, shall be added. The flask shall be attached to a reflux condenser by means of a ground glass joint, and heated on a steam bath, or other suitable heat source, until a precipitate is formed and for ten minutes after. Fifteen ml. of distilled water shall be added to dissolve the precipitate, and the solution refluxed gently on a steam bath for 45 to 60 minutes. The condenser and ground glass joints shall be washed

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with approximately 25 ml. of distilled water. The solution shall be cooled to room temperature, 10 drops of phenolphthalein indicator solution added, and the excess hydroxide titrated with 0.3N hydrochloric acid. A blank shall be run concurrently with the sample. The percentage of Di-2-ethylhexylphthalat shall be calculated as follows:

$$\text{Percent Di-2-ethylhexylphthalate} = \frac{19.53\ (A-B)N}{W}$$

where:

A = ml. of acid used to titrate the blank.

B = ml. of acid used to titrate the sample.

N = normality of acid.

W = weight of sample.

4.3.2 Determination of specific gravity at 20/20°C. -The specific gravity shall be determined at 20/20°C. in accordance with Standard MIL-STD-286 method 510.1.

4.3.3 Determination of acidity. -One hundred ml. of ethyl alcohol shall be transferred to a 250 ml. Erlenmeyer flask, three drops of phenolphthalein indicator and 5.9 ml. of 0.01N sodium hydroxide shall be added to the alcohol. When the 50 ml. portion of the sample is added to the flask and mixed, the solution must remain pink. The percent acid may be calculated as follows:

$$\text{Acidity percent} = \frac{8.3\ AN}{W}$$

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A = ml. of sodium hydroxide.

N = normality of sodium hydroxide.

W = weight of sample.

4.3.4 Determination of color. -

4.3.4.1 Preparation of color standard. -The stock color standard, containing 500 p.p.m. of platinic cobalt chloride (American Public Health Association color standard), (APHA) may be purchased (see 6.2), or may be prepared as follows:

A 1.245 gm. portion of potassium chloroplatinate (K_2PtCl_6 containing 0.500 gm. Platinum) and 1.000 gm. of crystallized cobaltous chloride ($CoCl_2$-$6H_20$, containing 0.248 gm. of cobalt) shall be dissolved in distilled water containing 100 ml. reagent grade hydrochloric acid. The solution shall be diluted to one liter with distilled water and mixed well. The solution has a color intensity equivalent to 500 p.p.m. of platinic cobalt chloride.

4.3.4.1.1 Calculation. -To prepare standards having a desired color, the ml. of 500 p.p.m. solution required for a given color standard shall be calculated by means of the following equation. This volume of standard solution shall be diluted with distilled water to the mark in a Nessler tube and mixed well.

$$\text{Volume of stock solution required} = \frac{AV}{500}$$

where:

A = Concentration in p.p.m. desired.

V = Final volume of diluted solution.

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4.3.4.2 Determination of color as received. -By means of a pipette, 10 mls. of the 500 p.p.m. standard shall be transferred to a 100 ml. Nessler tube, and diluted to the mark with distilled water. A second Nessler tube shall be filled to the mark with the sample as received. The intensity of the sample shall be compared with the standard by looking vertically downward through the two solutions against a white background. The sample color shall not be darker than the standard.

4.3.4.3 Determination of heat test color. -A Nessler tube shall be filled to the mark with the sample, placed in an oil bath at 220° ± 5 °C. and heated for 2 hours. Time shall be measured from the entry of the tube into the bath. The tube and contents shall be removed from the heating bath and allowed to stand at room temperature for 30 minutes. The sample shall then be compared with a color standard corresponding to 75 p.p.m. of platinum cobalt standard in the manner specified in 4.3.4.2. The sample color shall not be darker than the standard. (see 6.3)

4.3.5 Ash content. -A 10 gm. portion of the sample shall be transferred to an accurately tared porcelain crucible, evaporated to dryness on a hot plate, and the residue ignited to constant weight at red heat. The crucible and contents shall be cooled in a desiccator, and weighed. The increase in weight of the crucible shall be calculated to percentage of ash as follows:

$$\text{Percentage ash} = \frac{(B-A)100}{W}$$

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where:

A = weight of crucible

B = weight of crucible and ignited residue

W = weight of specimen

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. -Unless otherwise specified in contract or purchase order, Di-2-ethylhexylphthalate shall be packed in 55 gallon metal drums in accordance with Specification PPP-D-729 (Type I or Type II).

5.1.2 Level B. -Di-2-ethylhexylphthalate shall be packed as specified in 5.1.1.

5.1.3 Level C. -Di-2-ethylhexylphthalate shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Containers Specifications for rail shipments or National Motor Freight Rules and Container Specifications for truck shipments, as applicable.

5.2 Marking. -Unless otherwise specified in the contract or order, each container shall be plainly marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. -Procurement documents should specify the title, number and date of this specification.

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6.2 The stock color standard, containing 500 p.p.m. of platinic cobalt chloride (APHA color standard), a product of Hartman Laddon Company, Philadelphia, Pennsylvania, has been found satisfactory for use in the color determination (see 4.3.4).

6.3 Tubes made from pyrex or heat resistant glass should be used. (see 4.3.4.3).

Notice. -When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8270

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